# Can We See John Stockton's Heart Rate in the Box Score?

3-Post NBA Durability Series — Status 2026-10-18 (v7)

## The Premise

John Stockton and Boris Diaw reportedly share a resting heart rate of ~35 BPM — well below the elite-athlete norm of ~50 BPM. The question: does that physiology leave a detectable trace in box-score data as statistical resilience — fewer missed games, steadier late-game performance, more durable decision-making under fatigue? Stockton's durability is currently circulating on social media, providing a natural hook.

## Post Structure

| Post | Subject | Data Window |
|---|---|---|
| 1 | John Stockton + 14 PG comps | E1/E3 full career 1984–2003; E2/E4/E5 1996–2003 only |
| 2 | Boris Diaw + 5 PF comps (Nowitzki, Boozer, Jamison, Marion, Lewis) | 2004–2016, full BRef |
| 3 | All qualifying BRef-era guards + forwards + centers | 1980–2010 debuts |

Stockton ships first to ride the current conversation; Diaw follows immediately as independent methodological confirmation — different position, different era, full BRef coverage, no age-regime caveat. Post 3 addresses the cherry-picking objection: placing both players in the full distribution of all BRef-era players who meet the career-length threshold.

## Estimands

| | Metric | Data Tier | Status |
|---|---|---|---|
| **E1** | Games missed per 82 — career durability | Game logs | **Done — both cohorts** |
| **E3** | Shot selection: 3PA/FGA, FTA/FGA on B2B vs rest | Game logs | **Done — both cohorts** |
| **E2** | 4Q FT% degradation vs Q1–Q3 | Quarter box scores | **Done (PG) — Stockton +4.1 pp, only improver; PF pending** |
| **E4** | 4Q AST/TOV ratio vs full-game | Quarter box scores | **Degenerate at quarter resolution (52% zero-TOV rows); dropped** |
| **E5** | 4Q TS% vs Q1–Q3 | Quarter box scores | **Degenerate (21% coverage per quarter); dropped** |

| | Metric | Data Tier | Status |
|---|---|---|---|
| **E6** | Full Q1→Q2→Q3→Q4 scoring profile | Quarter box scores | **Done — both cohorts** |

E1 and E3 are primary. E6 (profile plot) generated from existing quarter splits; committed to both posts.

### Inference Approach

E1's denominator corrects for lockout-shortened seasons: 1999→50 games, 2012→66, 2020→65, 2021→72.

**n_treated = 1** makes bootstrap CIs invalid. For E1/E3: resample the treatment label across all N cohort players 5,000 times; report where the observed gap falls in the null distribution. For E2/E4/E5: mixed-effects model `metric ~ quarter * is_treated + rest_days + (1|player_id)` via `statsmodels.MixedLM`; permutation resamples `is_treated` at the player level (not game level) to preserve within-player correlation.

### Current Status

**Infrastructure: complete.** 141 tests passing across five scripts:

| Script | Purpose |
|---|---|
| `fetch_bref_data.py` | BRef game log scraper; HTML-cached per (player, season); 4 sec/request |
| `analyze_durability.py` | E1 + E3 + E7; permutation tests; multi-threshold (400/600/820 games, 5,000 TSA) |
| `fetch_quarter_splits.py` | Quarter box score scraper (named cohorts); cached by game ID; 3.5s/request + jitter |
| `analyze_quarter_splits.py` | E2/E4/E5; mixed-effects model; player-level permutation |
| `fetch_season_splits.py` | Season-level splits scraper (population); cached per (player, season); 4s/request |

**Data fetch: complete for all cohorts.**

| Status | Players |
|---|---|
| Full PG game logs + quarter splits | Stockton, Payton, Jackson, Harper |
| Full PG game logs only | Thomas, Hardaway, K. Johnson, Porter, Bogues, Strickland, Kidd, Nash, Blaylock, Anderson, Douglas |
| PF game logs + quarter splits (all 6) | Diaw, Nowitzki, Boozer, Jamison, Marion, Lewis |
| Population game logs (complete) | **2,067/2,067 players, 656,839 rows** — 0% NULLs |
| Population season splits (complete) | **738/738 players ( 400 games), 665,598 rows** — home/away, rest, W/L, month, opponent |

**Analysis: E7 (career TS%) implemented; population E1/E3 not yet run.**

| Estimand | Primary cohort | Sensitivity cohorts | Status |
|---|---|---|---|
| E7 TS% | 5,000 TSA (464 players, BRef standard) | 400 games (738), 820 games (233) | **Done** — Stockton rank 10/464 (p=0.022), robust across all sensitivities |
| E1/E3 population | 400 games (738 players, BRef standard) | 600 games (478), 820 games (233) | **Not yet run** |

**Post drafts:**

| Post | E1/E3 | E2/E4/E5 | E6 | E7 | Status |
|---|---|---|---|---|---|
| Stockton | Written | E2 written; E4/E5 dropped | Written | — | **Published** |
| Diaw | Written | E2 written (null); E4/E5 dropped | Written | — | **Published** |
| Population | Not run | — | — | Done | Outline only |

### Caveats (mandatory in every post)

1. **Selection-on-outcome bias** — 400-game threshold (BRef qualifying minimum) conditions on the dependent variable; E1 estimates durability *among players who completed substantial careers*, not all players.
2. **B2B state selection** — playing a B2B selects healthier player-states; within-player pairing removes stable effects but not selection.
3. **Blowout/garbage time** — starters sit in non-competitive games; 4Q minutes are not random. Compresses treatment effect toward zero.
4. **Intentional fouling** — close-game FTA inflation in Q4; FT% estimates partly reflect pressure context, not just physical state.
5. **Timeout-adjusted rest** — unmeasured confounder; coaches use Q4 timeouts strategically.
6. **Age-regime** — Stockton's quarter data covers ages 34–40 (career decline). A null within-game result does not contradict a prime-years resilience claim.
7. **Pre-1996 quarter gap** — BRef play-by-play starts 1996-97; only Stockton ages 34–40 are covered. Justin Jacobs (Squared Statistics) has video-reconstructed pre-1996 quarter logs that would extend the window to his prime (ages 22–33). Outreach drafted, not sent.
8. **No causality** — frame consistently as "consistent with" and "does not contradict."

### What's Left

| Step | Blocker | Est. |
|---|---|---|
| ~~Run final fetch audit, save `docs/fetch-audit.md`~~ | **Done** — 0 missing players; `docs/fetch-audit.md` written 2026-10-16 | — |

| Step | Blocker | Est. |
|---|---|---|
| ~~E7 career TS% analysis~~ | **Done** — Stockton rank 10/464 (98.1 pctile, p=0.022); 3-threshold sensitivity; figures in `data/figures/population*/` | — |
| ~~Season splits fetch~~ | **Done** — 738/738 players, 665,598 rows, `data/season_splits.parquet` | — |
| **Commit all pending** | `analyze_durability.py`, `fetch_season_splits.py`, `season_splits.parquet`, E7 CSVs + figures | <30 min |
| **Push `publish.yml` → verify CI** | 7 commits ahead of origin; blog-source push triggers two-deploy | <1 hr |
| **Run population E1/E3** at 400/600/820 thresholds | `python analyze_durability.py --cohort population` | 30 min |
| **Draft Post 3 outline** | E1/E3/E7 results; BRef disambiguation methods note | 2 hr |
| Season splits analysis script | home/away, rest, W/L angles → Post 4 | 1 day |
| E8: quarter splits for top-50 most-durable players | ~44 hrs fetch; deferred to Post 4 | future |
| Jacobs outreach (#22) — manual browser form | Human action required | 20 min |

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**NOTE:** Causal language forbidden. Null results are valid findings — state them plainly. Sourcing asymmetry: Diaw's 35 BPM comes from documented post-retirement interviews; the Stockton figure is less sourced — acknowledge this explicitly in the Stockton post.

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## Changelog

### v7 — 2026-10-18

- **Season splits fetch complete** — `fetch_season_splits.py` (new script); 738/738 400-game players; 665,598 rows; `data/season_splits.parquet` (6.8MB); captures home/away, rest, W/L, month, role, minutes, opponent splits per (player, season); quarter/half splits confirmed absent from BRef season splits pages (PBP-only)
- **E7 career shooting efficiency implemented** — `analyze_durability.py` extended; primary: 5,000 TSA (464 players, BRef TS% standard); Stockton rank 10/464 (98.1 pctile, p=0.022), robust across 400-game (rank 12/738, p=0.016) and 820-game (rank 10/233, p=0.043) sensitivities; Diaw 61.9 pctile primary, drops to 50.6 pctile among 10-year veterans
- **Population E1/E3 threshold standardized** — primary cohort now 400 games (BRef qualifying minimum); sensitivities at 600 and 820; `MIN_GAMES = 400` constant; outputs routed to `population/`, `population_600/`, `population_820/` subdirs
- **E8 scoped** — Q4 vs Q1 TS% margin for population requires per-game PBP (not season splits); deferred; top-50 most-durable players via `fetch_quarter_splits.py` is the feasible path (~44 hrs)

- **Threshold grounding**: 400 games = BRef rate stat qualifier (citable); 5,000 TSA = BRef TS% qualifier (citable); 820 games = NBA CBA 10-year veteran

**v6 — 2026-10-15/16**

- **Population fetch complete: 2,067/2,067 players, 656,839 rows** — all cohort members fetched including full HOF roster
- **BRef player ID disambiguation bug fixed** (see `docs/bref-id-disambiguation.md`):
  - Bug 1: 16 wrong IDs from name-collision clobbering in `build_population_cohort()`; all corrected via cached index HTML
  - Bug 2: ~59 HOFers missing due to BRef's `*` asterisk on HOF anchor text; scraper now strips `*` before lookup
  - Fix: replaced page-level `name_to_pid` dict with row-aligned `<th><a>` extraction; `drop_duplicates` → assertion
  - 95 contaminated rows purged (wrong Michael Smith + wrong Marcus Williams)
  - Cohort rebuilt from cache: 1,995 → 2,067 players (+72)
  - Issues filed: blog-source #2 (wrong IDs), #3 (HOF asterisk)
- **Off-by-one fix** (`fetch_bref_data.py` line 391): `debut_year + 1` → `debut_year`; was skipping every player's rookie season
- **Wave 3 local fetch**: ran on local IP (no VPN) resolving all BRef 403s that blocked Apollo; recovered 393/450 previously missing players
- **James Worthy**: fetched directly (926 rows, 12 seasons); last HOFer to land
- **3 confirmed empty pages**: Zevenbergen, Zimmerman, Zoet — all 1-game players, fetched but returned 17 rows combined; cohort complete
- `audit_fetch.py` written; `data/fetch_gaps.csv` generated; `docs/fetch-audit-wave1.md` saved as before baseline
- **Final fetch audit complete** — 2,067/2,067 players confirmed present; 0 missing players; `docs/fetch-audit.md` written; verdict FETCH COMPLETE

**v5 — 2026-10-14/15**

- **Population Wave 1 complete** — Docker container exited cleanly (exit code 0); 589,793 rows, 15M parquet pulled from Apollo
- **Fetch audit written** (`audit_fetch.py`) — cross-references `game_logs.parquet` vs `population_cohort.csv`; reports missing players, missing seasons, low game counts, NULL rates; writes `data/fetch_gaps.csv`
- **Wave 1 audit results** (see `docs/fetch-audit-wave1.md`):
  - 1,552/1,995 players fetched; 450 completely missing; 1,534 with 1 missing season
  - 2,375 missing (player, season) pairs; 1,228 low-game-count pairs (flagged, not errors)
  - 0% NULLs on all key columns — fetched rows are clean
  - Verdict: NEEDS SECOND WAVE
- **Off-by-one bug fixed** (`fetch_bref_data.py` line 391) — `first = int(row["debut_year"]) + 1` incorrectly skipped every player's rookie season; removed `+ 1`; the `debut_year` column already stores season-ending year consistent with BRef
- **DNS/VPN instability** caused the 450 fully-missing players — all requests exhausted retries during gluetun outages; Wave 2 will re-attempt via `fetch_gaps.csv`
- `docs/fetch-audit-wave1.md` created as before/after baseline for population post
- `docs/README.md` created — maps both wave audit files and explains naming convention

**v4 — 2026-10-13**

- **Apollo fetch: Docker production hardening** — `fetch_bref_data.py` rewritten with:
  - Env-configurable rate limits: `SESSION_COOLDOWN_PAGES=40` (was 20), `SESSION_COOLDOWN_SEC=120` (was 300)
  - `_rotate_vpn()`: calls Gluetun HTTP control server (`PUT /v1/vpn/status` stop → start) on consecutive 403s; 15s reconnect wait
  - 5-retry loop in `_get_html()` catching `Timeout` + `ConnectionError` (15*2^n backoff per attempt); previously no try/except existed
  - 403 counter → automatic VPN rotation after `MAX_403_BEFORE_ROTATE=3` consecutive blocks
  - Per-player `try/except Exception` in `fetch_population_cohort` — bad player skipped, run continues
- **`docker-compose.yml` updated**: `restart: on-failure:10` for `bref-fetch`; env vars for all rate-limit/VPN params; `gluetun-auth.toml` volume-mounted for Gluetun control server auth (v3.40+ requires auth config)
- **`gluetun-auth.toml` created**: grants unauthenticated access to `PUT/GET /v1/vpn/status` (control-only; no admin)
- **Containers rebuilt and running**: Gluetun + bref-fetch healthy; resumed from 594/1,995 players
- **Observed SSL EOF errors**: BRef resets SSL connections on VPN IPs (harder block than 403); correctly caught by `ConnectionError` handler and retried
- **Blog infrastructure: panel synthesis C1–C9** — Makefile rewrite (freeze-safe `clean`, `clean-freeze CONFIRM=yes`, `bootstrap`, `lint`, `lint-fix`, `check`); CI adds lint + pytest steps; JS canvas hardening (reduced-motion bail-out, DPR scaling, visibility-change pause); archive exclusions in `_quarto.yml`; `pyproject.toml` glob fix (`**` for nested test dirs); `.gitignore` excludes `game_logs.parquet`; changes staged, not yet committed

**v3.1 — 2026-10-12**

- **Docker setup for Apollo fetch** — migrated population fetch from nohup to Docker Compose with Gluetun ProtonVPN for IP rotation; `Dockerfile` + `docker-compose.yml` + `requirements.txt` (pinned to working venv: beautifulsoup4, lxml, numpy, pandas, pyarrow, requests)
- **Fixed missing deps**: first build failed with `ImportError: Unable to find a usable engine` (missing pyarrow); second failed with `ImportError: Import lxml failed`; both added to `requirements.txt`
- **Fetch resumed at 594/1,995 players** (236,972 rows) after Docker migration
- **403 blocks from ProtonVPN IPs**: BRef actively blocks datacenter/VPN IPs with HTTP 403; `on-failure:10` not yet set (set in v4)

**v3 — 2026-10-11**

- **E6 plots added** — `plot_e6_quarter_profile()` added to `analyze_quarter_splits.py`; generates Q1→Q4 FT% line chart highlighting treated player. PG: Stockton is the only player with a monotonic Q1→Q4 rise (76.9% → 80.6% → 83.2% → 84.7%). PF: Diaw shows slight Q4 decline (67.7%), Marion is the Q4 outlier (82.2%).
- **E6 figures generated** for both cohorts (`data/figures/pg/e6_quarter_profile.png`, `data/figures/pf/e6_quarter_profile.png`); figures committed.

- **E6 section added to both posts** — Stockton post: updated from "pending" placeholder to real figure with caption; conclusion updated with monotonic-increase note. Diaw post: new E6 section inserted after E5.
- **Diaw E2/E4/E5 complete** — PF quarter splits were already fetched (Oct 10); analysis complete; post draft finished.
- **Posts 1 + 2 published** — `draft: true` removed from Stockton and Diaw posts; `jupyter: blog` added to frontmatter; `_freeze/` outputs generated locally and committed; CI will serve frozen outputs without re-executing Python.
- **CI fixed** — kernel registration (`python -m ipykernel install --user --name blog`) added to `publish.yml`; last green run 2026-10-09.
- **Population fetch running locally** — PID 39001; ~109 population players fetched so far; ETA ~Oct 14.
- 141 tests passing (was 131 after Oct 10 session).

**v2 — 2026-10-10 (updated same day)**

- BRef ID fixes: Bogues (`boguemu01`), Harper (`harpede01`), Strickland (`stricro02`) — PG cohort grew from 12 → 15 players
- E1 + E3 analysis complete for both PG (15 players) and PF (6 players)
- Stockton post drafted with E1/E3 results; E2/E4/E5 sections stubbed pending figures
- Diaw post drafted with E1/E3 results; E2/E4/E5 sections stubbed
- Quarter splits fetched for Payton, Jackson, Harper, Stockton (5,585 rows)
- E2/E4/E5 analysis run for PG cohort; E2 interpretable (Stockton Q4 FT% +4.1 pp, only improver, p=0.494 at floor); E4/E5 dropped — degenerate at per-quarter resolution (52% zero-TOV rows; 21% TS% coverage)
- Stockton post now complete draft (E1/E3/E2 written; E4/E5 explained as methodologically degenerate)
- Population fetch scope expanded to include centers; running on Apollo (232/1,995 players, 99,883 rows)
- Population cohort size: 1,995 players (guards + forwards + centers, debuts 1980–2010)
- Fixed 5 bugs in `fetch_quarter_splits.py`: Age column KeyError, rate limit (0.35s→3.5s+jitter), 429 backoff, game_id Tm→Team column, season range (1996→1997)
- Fixed parallel `--player` race condition in `fetch_bref_data.py` with `fcntl.flock()`

**v1 — 2026-10-09**

- Initial status document; infrastructure complete (131 tests); data fetch incomplete